

ArcadeDB: One ACID Engine for Documents, Graphs, Vectors, and Time Series

[TODO: working title; alternatives in .notes strategy doc]

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Abstract—[TODO: Written last. Spine: multi-model unification as a real, adopted system, one page/WAL/MVCC-transaction pipeline under documents, graphs, key-value, time series, and vectors; every index type (LSM, full-text, geo, hash, dense/sparse vector) commits inside the same transaction; Raft replication ships model-agnostic WAL page diffs; one jar runs embedded in-process or as a server speaking six wire protocols. Deep dive: what adding the vector model to a unified substrate takes, including the deliberate transactional-source-of-truth / eventually-consistent-ANN-graph tradeoff. Evaluation vs specialist engines + lessons from a decade of multi-model engineering (OrientDB lineage).]

Index Terms—multi-model databases, ACID transactions, vector search, graph databases, embedded databases

I. INTRODUCTION

[TODO: Vector-era motivation: apps hold documents + relationships + embeddings of the same entities; composed specialist stacks pay consistency/ops glue. Thesis: unification at the storage layer, not the API gateway. Claims 1–4 from the audit. Explicitly claim integration, not new algorithms.]

II. BACKGROUND AND DESIGN GOALS

[TODO: PROSE PASS NEEDED. Raw co-author input below (Garulli, 2026-07-08).]

ArcadeDB descends from OrientDB, and its design goals are best understood as deliberate corrections of three OrientDB limitations its authors lived with:

- **Durability.** OrientDB’s write-ahead log was asynchronous, single-threaded, and (in the authors’ assessment) unreliable. ArcadeDB’s journal is synchronous, parallel, and crash-proof. It follows the point-of-no-return discipline detailed in §III.
- **Edge storage.** OrientDB managed edges with a tree-bag structure; ArcadeDB replaces it with a LIFO linked list that scales substantially better at high edge counts.
- **High availability.** The OrientDB multi-master model “simply doesn’t work” (authors’ words); ArcadeDB abandons it for a Raft leader/follower architecture (§VI).

[TODO: Design goals framing; what industry usage demanded. Third-party academic adoption: DatAasee [1] (meta-data catalog for research libraries) and AptBacterialDB [2] (ArcadeDB/openCypher graph layer for an antibacterial-aptamer database), both independent multi-model graph uses.

Christian’s “why we chose it” paragraph goes here or in Lessons.]

III. ARCHITECTURE: ONE SUBSTRATE, FIVE MODELS

ArcadeDB is organized around a single storage abstraction rather than a federation of per-model engines. Everything that persists, whether it holds documents, graph adjacency, key-value pairs, time-series points, or vector index segments, is a *paginated component*: a file of fixed-size pages managed by one page manager, modified through one transaction context, and made durable by one write-ahead log. The data models differ in the record types and index structures layered on top; they do not differ in how bytes reach disk, how changes become durable, or how replicas stay consistent. This section walks that pipeline bottom-up and states precisely which guarantees each model inherits from it.

A. Pages, Transactions, and the Log

All reads and writes go through page-level copy-on-write. A transaction that modifies any record obtains private copies of the affected pages from the page manager and accumulates its changes there; concurrent readers continue to see the unmodified originals. Commit is optimistic: at first phase, the engine validates that the pages read by the transaction have not been changed by a concurrent committer (a per-page version check), then serializes every dirty page, *including index pages*, into a single WAL entry. There is no separate index log and no index-specific recovery path: a secondary-index update, a graph edge insertion, and a document field change that happen in the same transaction land in the same log record and either all survive a crash or none do. Recovery replays the WAL forward from the last checkpoint; we measure recovery time and post-crash integrity under `kill -9` in §VII.

Durability is configurable per deployment in one dimension only: when the WAL buffer reaches the operating system (asynchronously on a background flusher by default, or `fsync` per commit under the strictest setting). We flag this knob here because it moves throughput by more than an order of magnitude and is therefore the first thing our evaluation pins when comparing against engines with different default contracts (§VII).

B. One Index Family Tree

Table ?? inventories the index and storage structures. The workhorse is an LSM tree [3] implemented as a paginated component: mutable in-memory pages absorb writes, immutable compacted segments serve reads, and compaction is an ordinary paginated write path. The full-text index is a thin analyzer layer over the same LSM tree (tokens as keys), and the geospatial index likewise reduces to LSM entries over space-filling-curve keys. The practical consequence of this reuse is uniform transactional behavior: because every index is a paginated component, the commit path of §III-A covers all of them with no per-index machinery. The two vector index families are the newest members of the tree and the most demanding test of the claim that the substrate generalizes; they get their own section (§IV).

C. Graph Storage, Stated Honestly

Vertices and edges are records in buckets, and adjacency is maintained as RID-linked lists attached to each vertex, a design inherited from OrientDB [4] and refined (the tree-bag edge container was replaced by a LIFO linked list that behaves better at high fan-out, §II). Edges without properties are stored as *light edges*: pure RID pairs in the adjacency structure with no edge record at all, which roughly halves the storage and I/O cost of plain connections. Traversal is pointer-chasing over the record store through the same page cache as every other access. We do not claim parity with purpose-built native graph storage on every workload; we claim, and measure in §VII, that RID-linked adjacency over a shared substrate delivers competitive point-traversal latency while inheriting transactions, recovery, and replication that a bolted-on graph layer would have to re-implement.

D. Replication Is Model-Agnostic

Because every model’s changes reduce to WAL’d page deltas, replication does not know what a document, an edge, or a vector is. The high-availability module (§VI) ships transaction page deltas through a Raft log [5]; followers apply them through the same recovery code path used after a crash. Adding a new data model to ArcadeDB, as §IV recounts for vectors, required *no* replication work: a model whose writes are paginated is replicated by construction. The one deliberate exception, derived search structures that are rebuilt rather than replicated, is discussed in §IV.

IV. CASE STUDY: ADDING THE VECTOR MODEL

Vector search is the sharpest recent test of the unification thesis: it arrived as a workload after the substrate was fixed, it is dominated by specialist engines with no transactional obligations, and it stresses exactly the properties (index build cost, memory, approximate results) that a shared substrate might be suspected of handling poorly. This section describes what adding dense and sparse vector search to ArcadeDB actually took, in the spirit of integration case studies such as SingleStore-V [6] and Cosmos DB’s DiskANN integration [7], and states the one place where we deliberately relaxed the substrate’s guarantees.

A. Dense Vectors: Transactional Records, Derived Graph

Dense embeddings are ordinary record properties: they live in buckets, are WAL’d, versioned, and replicated like any other field. Approximate search is served by a graph index built with the JVector library [8], which combines an HNSW-style layer hierarchy [9] with Vamana-style graphs within each layer [10]. Here we made the one deliberate exception to full unification: the ANN graph itself is an asynchronously maintained, WAL-disabled structure. Vector *records* are the transactional source of truth; the search graph is derived state, rebuilt from records if lost, and followers rebuild rather than receive it. The alternative, WAL-logging every neighbor-list mutation of an incremental graph build, would have coupled commit latency of unrelated transactions to ANN maintenance. The consequence is a bounded staleness window between a committed insert and its visibility in approximate search (newly inserted vectors are meanwhile served exactly from a delta buffer), which we consider the correct trade for an operational engine and state openly rather than claim a fully transactional ANN index. One design decision this enabled: because vectors-as-records are the ground truth, index parameters can be changed and the graph rebuilt without touching data, which we used in §VII to match graph degrees fairly across engines.

B. Sparse Vectors: An Inverted Index That Commits

Learned sparse retrieval (SPLADE-style [11]) maps text to high-dimensional sparse weight vectors and is conventionally served by standalone inverted-index engines with block-max pruning [12]–[14]. To our knowledge ArcadeDB is the first general-purpose transactional DBMS to ship this workload natively: `LSM_SPARSE_VECTOR` stores posting lists as LSM segments (the same paginated component family as every other index), with RID-delta compression, per-block weight quantization to int8, block maxima for WAND-style pruning at query time, and size-tiered compaction as the settle step. Because postings ride the LSM commit path, sparse index updates are transactional with document writes, a property none of the standalone engines offer. The quantization choice is a measured trade: int8 posting weights cost about half a point of recall@10 against exact scoring (§VII) and can be disabled per index (fp32 segments) where exact weights matter more than space.

C. What Integration Cost, Measured

Honesty about the bill: the substrate gave vectors durability, transactions, and replication for free, but not specialist performance for free. Three costs surfaced in our own measurements (§VII, §VIII): graph index construction at 10M vectors requires several times the raw vector size in heap because build-time structures hold vectors on-heap alongside JVector’s construction state; sparse query latency on real SPLADE term distributions degrades at corpus sizes where specialist engines do not; and settle steps (compaction) that specialists hide behind background services are visible operations here. We report these as findings with the same prominence as the wins,

and several became upstream fixes during the course of this work (§VIII).

D. A Note on Measurement Integrity

The sparse index’s original design documentation reported a $766\times$ speedup over brute-force scoring at 1M documents, a number that survived review because the same document also *retracted* an earlier $41,613\times$ claim traced to a broken baseline. We mention this not as trivia but as method: every headline number in this paper carries its measurement protocol, and where our own instruments were later found misleading (a memory metric that counted reclaimable page cache; a timeout that was our harness’s fault) the corrected number replaced the flattering one. §VII states the protocol once, and the raw results are published with the paper.

V. QUERY AND DEPLOYMENT SURFACES

[TODO: One executor + result model behind SQL/Cypher/GraphQL/Mongo (Gremlin via TinkerPop over the same storage); embedded in-process vs server; six wire protocols; cite the SciPy paper for Python binding internals.]

VI. HIGH AVAILABILITY

[TODO: Ratis-based Raft shipping shared-WAL page diffs =, model-agnostic replication; quorum/membership hardening lessons.]

VII. EVALUATION

[TODO: E1 sparse DB-vs-DB (Qdrant, Milvus) at 100k/1M/10M; E2 unified-ACID hybrid transaction vs composed stack incl. atomicity demonstration; E3 durability (kill -9) + Raft failover; E4 embedded vs server deployment axis; E5 dense sanity. Protocol: docker, pinned digests, identical cpuset/memory caps, N=5 mean $\pm$ std, serial re-runs for all reported numbers.]

VIII. LESSONS LEARNED AND TRADEOFFS

[TODO: The cost of unification (measured); where specialists win; single-node ceiling; async-ANN consequences; production war stories (co-authors).]

IX. RELATED WORK

[TODO: Multi-model (Cosmos DB API-level vs our storage-level; ArangoDB; Lu & Holubová survey; UniBench/M2Bench), vector DBs (Milvus, Manu, Qdrant, pgvector), sparse retrieval (SPLADE, BlockMax-WAND), embedded engines (SQLite, DuckDB).]

X. CONCLUSION

[TODO:]

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We thank Christian Himpe (University of Münster) for his contributions to ArcadeDB, for sharing his experience operating it in DatAasee [1], and for feedback on this manuscript. [TODO: AI-generated-content disclosure REQUIRED by ICDE 2027: name tools and affected content areas (drafting/editing assistance, experiment tooling). Keep updated as the paper evolves.]

REFERENCES

- [1] C. Himpe, “DatAasee – a metadata-lake as metadata catalog for a virtual data-lake,” *arXiv preprint arXiv:2409.05512*, 2024.
- [2] N. Bajiya, I. Gupta, and G. P. S. Raghava, “AptBacterialDB: A comprehensive, manually curated database of antibacterial aptamers,” *bioRxiv preprint*, 2026, doi:10.64898/2026.07.01.735956; uses ArcadeDB/openCypher for the aptamer–target knowledge graph.
- [3] P. O’Neil, E. Cheng, D. Gawlick, and E. O’Neil, “The log-structured merge-tree (LSM-tree),” *Acta Informatica*, vol. 33, no. 4, pp. 351–385, 1996.
- [4] D. Ritter, L. Dell’Aquila, A. Lomakin, and E. Tagliaferri, “OrientDB: A NoSQL, open source MMDMS,” in *Proceedings of the British International Conference on Databases (BICOD)*, 2021.
- [5] D. Ongaro and J. Ousterhout, “In search of an understandable consensus algorithm,” in *USENIX Annual Technical Conference*, 2014, pp. 305–319.
- [6] C. Chen, C. Jin, Y. Zhang, S. Podolsky, C. Wu, S.-P. Wang, E. Hanson, Z. Sun, R. Walzer, and J. Wang, “SingleStore-V: An integrated vector database system in SingleStore,” *Proceedings of the VLDB Endowment*, vol. 17, no. 12, pp. 3772–3785, 2024.
- [7] N. Upreti, H. V. Simhadri, H. S. Sundar, K. Sundaram *et al.*, “Cost-effective, low latency vector search with Azure Cosmos DB,” *Proceedings of the VLDB Endowment*, vol. 18, no. 12, pp. 5166–5183, 2025.
- [8] JVector, “JVector: A pure Java embedded vector search engine,” <https://github.com/datastax/jvector>, 2024.
- [9] Y. A. Malkov and D. A. Yashunin, “Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 4, pp. 824–836, 2020.
- [10] S. J. Subramanya, Devvrit, R. Kadekodi, R. Krishnaswamy, and H. V. Simhadri, “DiskANN: Fast accurate billion-point nearest neighbor search on a single node,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [11] T. Formal, B. Piwowarski, and S. Clinchant, “SPLADE: Sparse lexical and expansion model for first stage ranking,” in *Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021, pp. 2288–2292.
- [12] A. Z. Broder, D. Carmel, M. Herscovici, A. Soffer, and J. Zien, “Efficient query evaluation using a two-level retrieval process,” in *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2003, pp. 426–434.
- [13] S. Ding and T. Suel, “Faster top-k document retrieval using block-max indexes,” in *Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2011, pp. 993–1002.
- [14] N. Tonellotto, C. Macdonald, and I. Ounis, “Efficient query processing for scalable web search,” *Foundations and Trends in Information Retrieval*, vol. 12, no. 4–5, pp. 319–500, 2018.